AFC–7747

Sub. Code
1MMA4E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fourth Semester

Mathematics – Elective

ADVANCED STATISTICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Let a random sample of size 17 from the normal distribution $n(\mu,\sigma^2)$ yield $\overline{x}=4.7$ and $s^2=5.76$ determine a 90% confidence interval for μ.

2. Define biased and unbiased estimator.

3. Outline the principle of sufficient statistic.

4. Define complete family of p.d.f.

5. Explain unbiased minimum variance estimator.

6. Bringout the relationship between sufficient statistic and maximum likelihood estimator.

7. Explain most powerful test.

8. Define non-central t-distribution.

9. Show that the square of a non-central T random variable is a non- central F random variable.

10. Explain the relationship between testing of hypothesis and confidence interval.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Two independent random samples of sizes 10 and 7 respectively from two normal distributions $n(\mu_1,\sigma^2)$ and $n(\mu_2,\sigma^2)$ yield $\bar{x}=4.2$, $\bar{y}=3.4$, $s_1^2=49$, $s_2^2=32$. Find 90 confidence interval for $\mu_1-\mu_2$.

Or

(b) The observed value of the mean $\overline{X}$ of a random sample of size 20 from a normal distribution. $n(\mu,80)$ is 81.2. Find 90% confidence interval of μ.

12. (a) Prove that the nth order statistic of a random sample of size n form a uniform distribution with p.d.f. $f(x,\theta)=\begin{cases}\frac{1}{\theta}, & 0<x<\theta, \quad 0<\theta<\infty \\ 0, & elsewhere\end{cases}$ is a sufficient statistic for θ

Or

(b) Let $X_1,X_2,....X_n$ be a random sample of size n from a beta distribution with parameters $\alpha=\theta>0$ and $\beta=2$. Show that $\prod_{i=1}^{n} X_i$ is a sufficient statistic for θ.

13. (a) Discuss the Bayesian estimation.

Or

(b) Let S^2 be the variance of a random sample of size $n>1$ from a distribution that is $n(\mu,\theta)$. $0<\theta<\infty$. Prove that the cramer – Rao lower bound for the variance of the unbiased estimator $\frac{ns^2}{n-1}$ is $\frac{2\theta^2}{n}$.

14. (a) Let $X_1, X_2, ...X_{25}$ be a random sample of size 25 from a normal distribution $n(\theta,100)$. Find a uniformly most powerful critical region of size $\alpha=0.10$ for resting $H_o:\theta=75$ against $H:\theta>75$.

Or

(b) Let $X_1, X_2, ...X_{15}$ denote a random sample from a distribution that is $N(0,\theta)$ where the variance θ is a unknown positive number. Test the sample hypothesis $H_o:\theta=3$, against the alternate composite hypothesis $H_1:\theta>3$. Using uniformly most powerful test and obtain a best critical region of size 0.05.

15. (a) Define a non central χ^2 variate and obtain its p.d.f..

Or

(b) A random sample of size $n=6$ from a bivariate normal distribution yields the value of correlation coefficient to be 0.89. Test the hypothesis $H_0:p=0$ at 5% level of significance.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the Chi – square test of goodness of fit.

17. State and prove Rao – Blackwell theorem.

18. Let the random sample $x_1, x_2, ...x_n$ arise from $N(\theta_1, \theta_2)$. Then ffind $\hat{\theta}_1, \hat{\theta}_2$ of θ_1 and θ_2. Also find the covariance matrix.

19. Develop the procedure of sequential probability ratio test for testing $H_o : \theta = \theta_0$ against $H_1 : \theta = \theta_1$ for a distribution with p.d.f. $f(x;\theta)$.

20. Describe the analysis of variance for one – way classification.

AFC–7728

Sub. Code
1MMA1C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015.

First Semester

Mathematics

ANALYSIS — I

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Show that every neighborhood is an open set.

2. Define Cantor set. Give an example.

3. What is meant by complete metric space?

4. State ratio test.

5. Find the radius of convergence of the power series $\sum \frac{2^n z^n}{n^2}$.

6. If $\sum a_n$ converges absolutely, then prove that $\sum a_n$ converges.

7. What is the difference between continuity and uniform continuity?

8. Define the discontinuity of the second kind.

9. Examine for differentiability the function f given by

$$f(x) = \begin{cases} x \sin \frac{1}{x}, & x \neq 0 \\ 0 \quad , & x = 0 \end{cases}.$$

10. State mean value theorem.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Prove that A set E is open if and only if its complement is closed.

Or

(b) Show that compact subsets of metric spaces are closed.

12. (a) If $\overline{E}$ is the closure of a set E in a metric space X, then prove that $dia\ m\ \overline{E} = dia\ m\ E$.

Or

(b) Define e and prove that it is not rational

13. (a) If $\sum a_n = A$ and $\sum b_n = B$, then prove :

(i) $\sum (a_n + b_n) = A + B$

(ii) $\sum ca_n = CA$ for any fixed C.

Or

(b) If $\sum a_n$ is a series of complex numbers which converges absolutely, then prove that every rearrangement of $\sum a_n$ converges and they all converges to the same.

14. (a) If f is a continuous mapping of a metric space X into a metric space Y, and if E is a connected subset of X, then prove that $f(E)$ is connected.

Or

(b) Show that monotonic functions have no discontinuities of the second kind.

15. (a) Let f be defined for all real x and suppose that $|f(x) - f(y)| \leq (x - y)^2$ for all real x and y. Prove that f is constant.

Or

(b) Suppose f is a continuous mapping of [a, b] into R^K and f is differentiable in (a, b). Prove that there exists $x \in (a, b)$. Such that $|f(b) - f(a)| \leq (b - a)|f'(x)|$.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Let $\{E_n\}, n = 1, 2, 3,...$ be a sequence of countable sets and put $S = \bigcup_{n=1}^{\infty} E_n$. Prove that S is countable.

(b) Show that A subset E of the real line R' is connected if and only if it has the following property : If $x \in E, y \in E$ and $x < z < y$, then $z \in E$.

17. Prove the following :

(a) If $p > 0$, then $\lim_{n\to\infty} \frac{1}{n^p} = 0$.

(b) If $p > 0$, then $\lim_{n\to\infty} \sqrt[n]{p} = 1$.

(c) If $|x| < 1$, then $\lim_{n\to\infty} x^n = 0$.

18. (a) Suppose :

(i) $|C_1| \geq |C_2| \geq |C_3| \geq ....$

(ii) $C_{2m-1} \geq 0, C_{2m} \leq 0 \ (m = 1,2,3,...)$.

(iii) $\lim_{n\to\infty} C_n = 0$. Prove that $\sum C_n$ converges.

(b) State and prove Merten's theorem.

19. Let f be a continuous mapping of a compact metric space x into a metric space y. Prove that f is uniformly continuous on x.

20. State and prove L'Hospital's rule.

AFC–7729

Sub. Code
1MMA1C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Mathematics

DIFFERENTIAL GEOMETRY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define the rectifying plane.

2. Define Arc length.

3. What is meant by tangent surface?

4. Define cylindrical helix.

5. Define a helicoid.

6. Find the orthogonal trajectories of the circle $r = a\cos\theta$.

7. Define geodesic curvature.

8. State whitehead theorem.

9. Define an elliptic point.

10. State Rodrigue formula.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Calculate the curvature and torsion of the cubic curve given by $\vec{r} = (u, u^2, u^3)$.

Or

(b) Show that $[\dot{\vec{r}}, \ddot{\vec{r}}, \dddot{\vec{r}}] = 0$ is a necessary and sufficient condition that the curve be plane.

12. (a) Derive the equation of the involute.

Or

(b) Write a brief note on Helices.

13. (a) Derive the expression for the sine and cosine of the angle between parametric curves.

Or

(b) Show that a metric is invariant under a parameter transformation.

14. (a) State and prove Liouville's theorem for kg.

Or

(b) Prove that if the orthogonal trajectories of the curves $v = \text{constant}$ are geodesics, then H^2/E is independent of u.

15. (a) Write a note on second fundamental form of a surface.

Or

(b) Write a short note on 'Dupin indicatrix'.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Derive Serret–Frenet formulae.

17. Show that the torsion of an involute of a curve is equal to $\frac{\rho(\sigma\rho' - \sigma'\rho)}{(\rho^2 + \sigma^2)(c - r)}$.

18. Discuss isometric correspondence in detail.

19. Prove that if (λ, μ) is the geodesic curvature vector, then $Kg = \frac{-H\lambda}{Fu' + Gv'} = \frac{H\mu}{Eu' + Fv'}$.

20. Show that a necessary and sufficient condition for a surface to be a developable is that its Gaussian curvature shall be zero.

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AFC–7730

Sub. Code
1MMA1C4

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Mathematics

DIFFERENTIAL EQUATIONS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. When a function is analytic at x_0?

2. State the existence theorem for initial value problem for homogeneous equation.

3. Find the singular points of the equation $x^2y'' + (x + x^2)y' - y = 0$ and determine whether they are regular singular points or not.

4. Find all solutions of the equation $x^2y'' + 2xy' - 6y = 0$ for $x > 0$.

5. Eliminate the arbitary function from $z = xy + f(x^2 + y^2)$.

6. Find a complete integral of the equation $(p + q)(z - xp - yq) = 1$.

7. Find the complementary function of the P.D.E. $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial y^2} = x - y$.

8. Prove that $F(D, D')e^{ax+by} = F(a, b)\, e^{ax+by}$.

9. Explain interior Neumann problem.

10. Write the Laplace equation in spherical polar coordinates.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) If $\phi_1, \phi_2, \ldots \phi_n$ are n solutions of $L(y) = y^{(n)} + a_1(x)\, y^{(n-1)} + \ldots + a_n(x)\, y = 0$ on an interval I, prove that they are linearly independent if and only if $W(\phi_1, \phi_2 \ldots \phi_n)(x) \neq 0$ for all x in I.

Or

(b) One solution of $x^3 y''' - 3x^2 y'' + 6xy' - 6y = 0$ for $x > 0$ is $\phi_1(x) = x$. Find a basis for the solution for $x > 0$.

12. (a) Find the singular points of the equation $(x^2 + x - 2)^2 y'' + 3(x + 2)\, y' + (x - 1)\, y = 0$ and determine whether they are regular singular points or not.

Or

(b) Compute the indicial polynomial and their roots of the equation $x^2 y'' + (x + x^2)\, y' - y = 0$.

13. (a) Explain Charpit's method of solving the partial differential equation $f(x, y, z, p, q) = 0$.

Or

(b) Find the general solution of the differential equation $x^2 \frac{\partial z}{\partial x} + y^2 \frac{\partial z}{\partial y} = (x + y) z$.

14. (a) Solve $r + s - 2t = e^{x+y}$.

Or

(b) Solve $\frac{\partial^4 z}{\partial x^4} + \frac{\partial^4 z}{\partial y^4} = 2 \frac{\partial^4 z}{\partial x^2 \partial y^2}$.

15. (a) Explain Longitudinal sound waves.

Or

(b) Find the dementary solution of Laplace equation.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Find all solutions of the equation $y'' - \frac{2}{x^2} y = x$, $0 < x < \infty$.

17. Find the solution of the Bessel equation $x^2 y'' + xy' + (x^2 - \alpha^2) y = 0$.

18. Find the complete integral of the following PDE.

(a) $p^2 z^2 + q^2 = 1$

(b) $p^2 y (1 + x^2) = q x^2$.

19. Reduce the equation $\frac{\partial^2 z}{\partial x^2} + 2\frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 0$ to canonical form and hence solve it.

20. Find the solution of the Laplace equation in cylindrical coordinates.

AFC–7731

Sub. Code
1MMA1E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Mathematics

Elective – MECHANICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define virtual work.

2. State the Konig's theorem.

3. Define velocity dependent potential.

4. What is meant by dissipative function?

5. Define cyclic co-ordinates.

6. State the two body central force problem.

7. State Bertrand's theorem.

8. How orbits are classified?

9. Write the formula for orbital radius.

10. State the kepler problem.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) State and prove conservation theorem for angular momentum of a particle.

Or

(b) Find differential equations of motion for a spherical pendulum of length l.

12. (a) Prove that the centripetal equation of a bead sliding on a uniform ratating wire in a force – free space is $\ddot{r} = rw^2$.

Or

(b) Discuss the application of Lagrange formulation.

13. (a) Obtain the minimum surface of revolution.

Or

(b) Using Hamitons equations of motion discuss the motion of a simple pendulum.

14. (a) Show that the energy conservation theorem can be written in the form $E = v' + \frac{1}{2}mv^2$.

Or

(b) State and prove virtual theorem.

15. (a) Obtain the solution of Kepler's problem.

Or

(b) Explain the concept of "stress may be given for problems in the text".

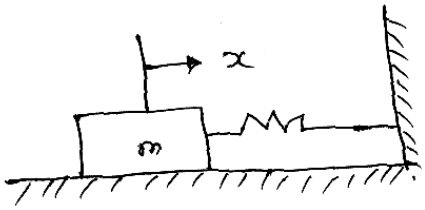
Part C (3 × 10 = 30)

Answer any **three** questions.

16. Derive Lagranges equation for holonomic constraints.

17. Discuss briefly about potentials.

18. Given a mass – spring system consisting of a mass m and linear spring of stiffness k as shown in fig. Find the equations of motion using the Hamiltonian procedure. Assume that the displacement is measured form the unstressed position of the spring.



19. Write briefly about the classification of orbits.

20. Show that the central force problem is solvable interms of elliptic functions when the for is a power law function of the distance with the following exponents $n = -\frac{3}{2}, \frac{-5}{2}, \frac{-1}{3}, \frac{-5}{3}, \frac{-7}{3}$.

AFC–7732

Sub. Code
1MMA2C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Mathematics

ALGEBRA – II

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define subspace of a vector space V.

2. Define linearly independent and linearly dependent vectors.

3. Define annibilator of a subspace.

4. Define inner product space.

5. Define algebraic extension of F.

6. If F is of characteristics $p \neq 0$, then prove that for $a,b \in F$ $(a+b)^{pm} = a^{pm} + b^{pm}$.

7. Define fixed field of G and give an example.

8. Define normal extension of F.

9. Define rank of the linear transformation.

10. Define Hermitian and normal linear transformation on V.

Part B $(5 \times 5 = 25)$

Answer **all** questions either (a) or (b).

11. (a) Prove that if V is the internal direct sum of $U_1, U_2 .... U_n$, then prove that V is isomorphic to the external direct sum of $U_1, U_2 .... U_n$.

Or

(b) If S, T are subsets of V, then prove that following

(i) $S \subset T \Rightarrow L(S) \subset L(T)$

(ii) $L(S \cup T) = L(S) + L(T)$.

12. (a) If W is a subspace of V, then prove that $A(A(W)) = A(W)$.

Or

(b) State and prove Schwanz inequality.

13. (a) State and prove the transitivity property of finite extensions.

Or

(b) Prove that a polynomial of degree n over a field can have atmost n roots in any extension field.

14. (a) Prove that $G(K, F)$ is a subgroup of all automorphisms of K.

Or

(b) Prove that if K is a field and if $\sigma_1, \sigma_2 ... \sigma_n$ are distribution automorphisms of K, then it is impossible to find elements $a_1, a_2 .... a_n$ not all zero is K such that $a_1 \sigma_1(u) + a_2 \sigma_2(u) + ... + a_n \sigma_n(u) = 0$ $\forall u \in K$.

15. (a) If V is finite dimensional over F, then prove that $T \in A(V)$ is regular $\Leftrightarrow$ T maps V onto V.

Or

(b) If $T \in A(V)$, then prove the following

(i) $T^* \in A(V)$

(ii) $(T^*)^* = T$.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. If V is finite dimensional and W is a subspace of V, then prove that W is finite dimensional, $\dim W \leq \dim V$ and $\dim(V/W) = \dim V - \dim W$.

17. Prove that if V and W are two vector spaces over F of dimensions m and n respectively, then Hom (V, M) is of dimension mn over F.

18. Prove that the element $a \in K$ is algebraic over F if and only if $F(a)$ is a finite extension of F.

19. Prove that K is a normal extension of F if and only if K is the splitting field of some polynomial over F.

20. Prove that the linear transformation T on V is unitary if and only if it takes an orthonmal basis of V into an orthonormal basis of V.

AFC–7733

Sub. Code
1MMA2C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Mathematics

ANALYSIS — II

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define Stieltje's integral.

2. Define refinement of a partition.

3. Define uniform convergence.

4. Find $\lim_{n\to\infty} n^2 x\left(1 - x^2\right)^n,\ 0 \le x \le 1,\ n = 1, 2, 3, \ldots$

5. Define Analytic function.

6. Define Orthogonal system of function.

7. Define the counting measure.

8. Define Borel measurable.

9. Define integrable function.

10. Define the lebesgae integral of f over a set E.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Prove that $\int_{\bar{a}}^{b} f\, d\alpha \le \int_{a}^{\bar{b}} f\, d\alpha$.

Or

(b) If f is monotonic on $[a, b]$ and if α is continuous on $[a, b]$ then prove that $f \in R(\alpha)$.

12. (a) Let $\{f_n\}$ be a sequence of functions defined on E and if $|f_n(x)| \le M_n$ $(x \in E, n = 1, 2, 3, \ldots)$ then prove that Σf_n converges uniformly on E. If ΣM_n converges.

Or

(b) If K is a compact metric space, if $f_n \in (K)$ for $n = 1, 2, 3, \ldots$ and if $\{f_n\}$ converges uniformly on K, then prove that $\{f_n\}$ is equicontinuous on K.

13. (a) Prove that $\int_0^1 t^{x-1}(1-t)^{y-1}\, dt = \dfrac{\Gamma(x)\,\Gamma(y)}{\Gamma(x+y)}$, if $x, y > 0$.

Or

(b) if ΣC_n converges, put $f(x) = \sum_{n=0}^{\infty} C_n x^n$, $(-1 < x < 1)$ then show that $\lim_{x \to 1} f(x) = \sum_{n=0}^{\infty} C_n$.

14. (a) If E_1 and E_2 are measurable then show that $E_1 \cup E_2$ is also measurable.

Or

(b) Show that the interval (a, ∞) is measurable.

15. (a) Let ϕ and ψ be simple functions which vanish outside a set of finite measure, then prove that $\int (a\phi + b\psi) = a\int \phi + b\int \psi$.

Or

(b) State and prove monotone convergence theorem.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. If γ' is continuous on $[a, b]$ then prove that γ is rectifiable and its length $\wedge(\gamma) = \int_a^b |\gamma'(t)|\, dt$.

17. If f is a continuous complex function on $[a, b]$, prove that there exists a sequence of polynomials p_n such that $\lim_{n\to\infty} p_n(x) = f(x)$.

18. State and prove Stirling's formula.

19. Prove that the outer measures of an interval in its length.

20. (a) State and prove Bounded convergence theorem.

(b) State and prove Fatou's lemma.

AFC–7734

Sub. Code
1MMA2C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Mathematics

NUMERICAL ANALYSIS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Find the critical points of $F(x) = x_1^3 + x_2^2 - 2x_1^2 + 3x_2^2 - 8$.

2. Find the spectral radius of the matrix $\begin{pmatrix} 3 & 2 \\ 1 & 4 \end{pmatrix}$.

3. Define Hermite polynomials $H_k(x)$ and find $H_2(x)$ and $H_3(x)$.

4. Show that, if $P(x)$ is a polynomial of degree less than K, then $P(x)$ is orthogonal to $P_k(x)$, for $K = 2$ only.

5. Find $\frac{dy}{dx}$ at $x = 50$ from the following data :

x :	50	51	52	53	54	55	56
y :	3.684	3.7084	3.7325	3.7563	3.7798	3.803	3.8259

6. Write down Simpson's $\frac{1}{3}$ formula and Simpson's $\frac{3}{8}$ formula for integration.

7. Find the general solution of the differential equation $\frac{d^3y}{dx^3} - 2\frac{d^2y}{dx^2} - \frac{dy}{dx} + 2y = 0$.

8. Find the general solution of the difference equation $y_{n+3} - 2y_{n+2} - y_{n+1} + 2y_n = 0$.

9. State the recursion formula for the Runge – Kutta method of order 2.

10. What do you mean by an initial value problem. Give an example.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Use Newton's method to solve the non – linear system $x^3 - 3xy^2 + 1 = 0$ and $3x^2y - y^3 = 0$ given $x_0 = y_0 = 1$.

Or

(b) Use the steepest descent method to find the maxima and minima of the function $F(x_1, x_2) = \frac{x_1^3}{3} + x_2^2 x_1 + 3$.

12. (a) Calculate a good polynomial approximation of degree n on $0 \le x \le 1$ to $f(x) = \sqrt{x}$ for $n = 1,2,3,4,5$.

Or

(b) Let the scalar product $< g, h > = \int_a^b g(x)\, h(x)\, w(x)\, dx$.

Then prove that $P_K(x)$ has K simple real zeroes, all of which lie in the interval (a,b).

13. (a) Derive the corrected trapezoidal rule and state its error.

Or

(b) A curve passes through the points (1, 2), (1.5, 2.4) (2, 2.7), (2.5, 2.8), (3, 3), (3.5, 2.6) and (4, 2.1). Obtain the area bounded by the curve, the $x-$axis, $x=1$ and $x=4$. Also, find the volume of solid of revolution got by revolving this area about the $x-$axis.

14. (a) By means of Taylor series expansion, find y at $x=0.1$ correct to three significant digits, given $\frac{dy}{dx}-2y=3e^x, y(0)=0$.

Or

(b) Find the general solution of the difference equation $y_{n+2}-4y_{n+1}+3y_n=2^n, y_0=0, y_1=1$.

15. (a) Solve using shooting method $y''=2y^3, y(1)=1$, $y(2)=1/2$, taking $y'(1)=0$ as a first guess.

Or

(b) Use finite difference method to solve for $y(x)$ given $\frac{d^2y}{dx^2}+y+1=0,\ x\in(0,1),\ \ y(0)=0,\ \ y(1)=0$ taking $h=1/4$.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Solve the non – linear system $x^2+xy^3=9, 3x^2y-y^3=4$ using the fixed point iteration method.

17. Using the appropriate recurrence relation, generate the first five Hermite polynomials. Also the polynomials orthogonal? Justify.

18. Derive the Simpson's rule together with its error term. Hence, solve $\int_0^6 \frac{dx}{1+x^2}$ taking $h = 0.5$.

19. Compute $y(0.2)$ given $\frac{dy}{dx} + y + xy^2 = 0, y(0) = 1$ by taking $h = 0.1$ using Runge – Kutta method of order 4 correct to four decimals.

20. Using Milne's predictor corrector method, find $y(0.6)$, (taking $h = 0.2$), for the equation $y' = y - x^2, y(0) = 1$, obtaining the starting values by Taylor's series method.

AFC–7735

Sub. Code
1MMA2C4

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Mathematics

PROBABILITY AND STATISTICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Let X have the p.d.f. $f(x)=\begin{cases}\frac{x}{6}, & x=1,2,3\\ 0 & elsewhere\end{cases}$. Determine $E(X^2)$.

2. Let X have the p.d.f. $f(x)=\begin{cases}\frac{1}{x^2}, & 1<x<\infty\\ 0 & elsewhere\end{cases}$. Show that the mean value of X does not exist.

3. Let the joint p.d.f. of x_1 and x_2 be $f(x_1,x_2)=12x_1x_2(1-x_2)\ 0<x_1<1, 0<x_2<1;$ zero elsewhere. Are x_1 and x_2 stochastically independent? Justify.

4. If X and Y have the joint p.d.f. $f(x,y)=\begin{cases}2 & 0<x<y<1\\ 0 & elsewhere\end{cases}$ obtain the marginal density functions.

5. If X is $n(2,25)$ calculate $\Pr(0 < x < 10)$.

6. X is a binomial variable with parameters $n = 7$ and $p = 1/2$. Find its moment generating function and $P(X = 5)$.

7. Let the p.d.f. $f(x) = \begin{cases} kx(3-x)^4, & 0 < x < 3 \\ 0, & elsewhere \end{cases}$. Determine the constant K.

8. Let X have the p.d.f. $f(x) = \begin{cases} x^2/9, & 0 < x < 3 \\ 0, & elsewhere \end{cases}$. Find the p.d.f. of $y = x^3$.

9. State any two theorems on limiting distributions.

10. Let $\overline{X}$ denote the mean of a random sample of size 100. From a distribution that is $x^2(50)$. Find an approximate value of $pr(49 < \overline{x} < 51)$.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) If X has the p.d.f. $f(x) = \begin{cases} \left(1/2\right)^x, & x = 1,2,3,..... \\ 0, & elsewhere \end{cases}$. Find the m.g.f. , mean and variance of X.

Or

(b) There are five red chips and three blue chips in a bowl. The red chips are numbered 1, 2, 3, 4, 5 respectively and the blue chips are numbered 1, 2, 3 respectively. If two chips are to be drawn at random and without replacement. Find the probability that these chips have either the same number or the same colour.

12. (a) Two random variables X and Y are connected by the equation $ax + by + c = 0$. Show that the correlation between X and Y is -1 if the sign of a and b are same and +1 if they are different.

Or

(b) Let $f(x_1, x_2) = \begin{cases} e^{-x_1, x_2,} & 0 < x_1 < \infty, 0 < x_2 < \infty \\ 0, & otherwise \end{cases}$ be the joint p.d.f of the random variables x_1 and x_2. Show that x_1 and x_2 are stochastically independent and that $E\left[e^t (x_1 + x_2)\right] = (1-t)^{-2}, t < 1$.

13. (a) Compute the measures of skewness and kurtosis of a gamma distribution with parameters α and β.

Or

(b) Let X have a poisson distribution with parameter $\mu > 0$. If K is a non-negative interger and if $\mu'_k = E(x^k)$, prove that $\mu'_{(k+1)} = \mu\left(\mu'_k + \frac{d\mu'_k}{d\mu}\right)$.

14. (a) Let X have the p.d.f. $f(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & elsewhere \end{cases}$. Prove that $y = 2\ln X$ has a chi-square distribution with 2 degrees of freedom.

Or

(b) Let X_1, X_2 be a random sample of size 2 from a distribution with p.d.f. $f(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & elsewhere \end{cases}$. Find the distribution function and p.d.f. of $Y = \frac{X_1}{X_2}$.

15. (a) Let Y_n be the nth order statistic of a random sample $X_1, X_2, \ldots X_n$ from a distribution with p.d.f.

$f(x) = \begin{cases} 1/\theta, & 0 < x < 1, \theta < y < \infty \\ 0, & elsewhere \end{cases}$. Find the limiting distribution on Y_n.

Or

(b) Let $S^2 n$ denote the variance of a random sample of size n from distribution which is $n_x(\mu, \sigma^2)$. Prove that $\frac{ns^2}{n-1}$ converges stochastically to σ^2.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) A random variable X has a p.d.f. $f(x) = \begin{cases} 3x^2, & 0 < x < 1 \\ 0 & elsewhere \end{cases}$. Find 'a' and 'b' such that $pr(x < a) = p(x > a)$ and $P(x > b) = 0.05$.

(b) State and Prove Baye's theorem.

17. (a) Prove that the correlation coefficient always lies between –1 to 1.

(b) Joint distribution of the random variables X and Y is $f(x, y) = 4xy, e^{-(x^2+y^2)}, x > 0, y > 0$. Examine whether X and Y are independent.

18. Show that Poisson distribution as the limiting form of binomial distribution.

19. Derive the p.d.f. of F-distribution and obtain its mode.

20. State and prove central limit theorem.

AFC–7737

Sub. Code
1MMA3C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Mathematics

COMPLEX ANALYSIS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Find the radius of convergence of the power series $\Sigma \frac{z^n}{n!}$.

2. If $g(w)$ and $f(z)$ are analytic functions, prove that $g(f(z))$ is also analytic.

3. Evaluate $\int_C \frac{z\,dz}{z^2-1}$ where C is the positively oriented circle $|z| = 2$.

4. If a and b are any two points lying in the same region determined by γ, prove that $n(\gamma, a) = n(\gamma, b)$.

5. Show that the function $\sin z$ has an essential singularity at ∞.

6. Prove that $f(z) = e^{1/z}$ has a removable singularity at $z = 0$.

7. Find the poles and residues of $f(z) = \tan z$ inside the circle $|z| = 2$.

8. Evaluate $\int_C \frac{dz}{2z+3}$ where C is $|z| = 2$.

9. State the Laurent series.

10. Obtain the expansion of $\frac{z-1}{z^2}$ in a Taylor series in powers of $(z-1)$.

Part B (5 × 5 = 25)

Answer **all** the questions, choosing either (a) or (b).

11. (a) Find the bilinear transformation which maps the prints $2, i, -2$ on to $1, i, -1$ respectively.

Or

(b) Show that an analytic function cannot have a constant absolute value without reducing to a constant.

12. (a) State and prove Schwarz lemma.

Or

(b) Obtain a necessary and sufficient condition for the line integral $\int_V pdx + qdy$ define in Ω to depend only on the end point of V.

13. (a) If $f(z)$ is analytic and non-constant in a region Ω, prove that its absolute value $|f(z)|$ has no maximum in Ω.

Or

(b) Prove that an analytic function comes arbitrarily close to any complex value in every neighbourhood of an essential singularity.

14. (a) State and prove Rouche's theorem.

Or

(b) Compute $\int_0^{\pi} \frac{d\theta}{17 - 8\cos\theta}$.

15. (a) Prove that for $|z| < 1$,

$$(1+z)(1+z^2)(1+z^4)(1+z^8)\cdots = \frac{1}{1-z}.$$

Or

(b) Show that the Laurent series development is unique.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) State and prove Luca's theorem.

(b) Prove that a linear transformation carries circles into circles.

17. If the function $f(z)$ is analytic on the rectangle R defined by the inequalities $a \le x \le b$, $c \le x \le d$, Prove that $\int_{\partial R} f(z)dz = 0$ where ∂R denote the boundary of R.

18. State and prove the local mapping theorem.

19. Evaluate

(a) $\int_0^{\infty} \frac{\cos x\, dx}{x^2 + a^2}$ where a is real.

(b) $\int_0^{\infty} \frac{x^2\, dx}{x^4 + 5x^2 + 6}$.

20. State and prove the Mittag-Leffler's theorem.

AFC–7738

Sub. Code
1MMA3C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Mathematics

TOPOLOGY – I

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Write down the conditions for a collection β of subsets of a nonempty set X to be a basis.

2. Is the subset $[a, b]$ of R closed. Justify.

3. Explain the term homeomorphism with an example.

4. State the sequence lemma.

5. Show that the space $Y = [-1, 0) \cup (0, 1]$ is disconnected in R.

6. Give an example of a connected space which is not path connected.

7. Justify that the real line R is not compact.

8. What is a Lebesgue number?

9. Prove that R_l is normal.

10. What do you mean by a completely regular space?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Let $\mathcal{B}$ and $\mathcal{B}'$ be bases for the topology J and J' respectively on X. Show that the following statements are equivalent.

    (i) J' is finer than J

    (ii) For each $x \in X$ and each basis element $B \in \mathcal{B}$ containing x, there is a basis element $B' \in \mathcal{B}'$ such that $x \in B' \subset B$.

Or

(b) Define subspace topology. For a subset Y of a topological space X, define $J_y = \{Y \cap U / U \in J\}$. Prove that J_y is a topology.

12. (a) Show that the topologies on R^n induced by the Euclidean metric d and the square metric P are the same as the product topology on R^n.

Or

(b) State and prove uniform limit theorem.

13. (a) Prove that a discrete space is disconnected and that the image of a connected space under a continuous map is connected.

Or

(b) State and prove intermediate value theorem.

14. (a) Show that every compact subset of a Hansdorff space is closed.

Or

(b) Prove that compactness implies limit point compactness, but not conversely.

15. (a) Define the following terms :

(i) Regular space

(ii) Normal space.

Or

(b) Show that a subspace of a regular space is regular and that a product of regular spaces is regular.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Let $\mathcal{B}$ be a basis for the topology of X and let $\mathcal{C}$ be a basis for the topology of Y. Show that the collection $D = \{B \times C \mid B \in \mathcal{B} \text{ and } C \in \mathcal{C}\}$ is a basis for the topology of $X \times Y$ by proving necessary lemma.

17. Let $\overline{d}(a,b) = \min\{|a-b|, 1\}$ be the standard bounded metric on R. If x and y are two points of R^w, define $D(x,y) = \text{lub}\left\{\frac{\overline{d}(x_i, y_i)}{i}\right\}$. Prove that D is a metric that induces the product topology on R^w.

18. Show that the Cartesian product of connected spaces is connected.

19. Prove that the product of finitely many compact spaces is compact.

20. State and prove Urysohn lemma.

AFC–7739

Sub. Code
1MMA3C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Mathematics

GRAPH THEORY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define an isomorphism between two graphs. Give an example.

2. Prove that the number of odd degree vertices is even.

3. Define a block of a graph with an example.

4. What is meant by an internally-disjoint paths?

5. Define a matching. Give an example.

6. What is meant by k-edge chromatic? Give an example.

7. Show that $r(k, l) = r(l, k)$.

8. Draw a 3-chromatic graph.

9. State Jordan curve theorem in the plane.

10. Define the dual of a plane graph.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b)

11. (a) Prove the following

(i) $\varepsilon\left(K_{m,n}\right) = mn$

(ii) If G is simple and tripartite, then $\varepsilon \le \frac{\gamma^2}{4}$.

Or

(b) If G is simple and $\varepsilon > \left(\frac{\gamma - 1}{2}\right)$ then prove that G is connected.

12. (a) With the usual notations, prove that $k \le k' \le \delta$.

Or

(b) Prove that $C(G)$ is well defined.

13. (a) State and prove Berge theorem in a matching.

Or

(b) Prove that the Petersen graph is 4-edge chromatic.

14. (a) With the usual notations, prove that

$r(k, l) \le \binom{k + l - 2}{k - 1}$.

Or

(b) Prove that every critical graph is a block.

15. (a) Show that K_5 is non planar graph.

Or

(b) If two bridges overlap, then prove that they are skew or else they are equivalent 3-bridges.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Prove that a graph is bipartite if and only if it contains no odd cycle.

17. State and prove the Chvatal theorem.

18. If G is simple graph then prove that either $\chi' = \Delta$ or $\chi' = \Delta + 1$.

19. State and prove Brook's theorem for vertex colorings.

20. (a) If G is a connected plane graph, then prove that $\gamma - \varepsilon + \phi = 2$.

(b) State five colour theorem.

(c) Discuss the four-colour conjecture.

AFC–7740

Sub. Code
1MMA3C4

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Mathematics

OPERATIONS RESEARCH

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define a directed loop.

2. Write any two rules for constructing the network.

3. What are the costs associated with inventory?

4. Define load time.

5. What are the limitations of queuing theory?

6. State any two applications of queuing theory in business enterprises.

7. What is meant by multichannel queuing model?

8. Draw the transition-rate diagram.

9. Narrate the dichotomous search method.

10. Write short note on separable programming problem.

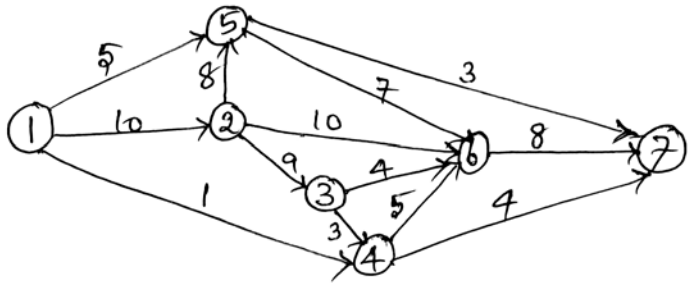
Part B (5 × 5 = 25)

Answer **all** questions by choosing either (a) or (b).

11. (a) Explain Floyd's algorithm.

Or

(b) Determine the critical path for the following project network.



12. (a) Enumerate the no-setup model.

Or

(b) Lube car specializes in fast automobile oil change the garage buys car oil in bulk at \$ 3 per gallon. A discount price of \$ 2.50 per gallon is available if lube car. Purchases more than 1000 gallons. The garage services approximately 150 cars per day and each oil change takes 1.25 gallons. Lube car stores bulk oil at the cost of \$0.02 per gallon per day. Also, the cost of placing an order for bulk oil is \$20. There is a 2-day lead time for delivery. Determine the optimal inventory policy.

13. (a) What is a queue? Give an illustration. Also describe the basic elements of a queuing model.

Or

(b) The florist section in a grocery store stocks 18 dozen roses at the beginning of each week. On the average, the florist sells 3 dozens a day (one dozen at a time), but the actual demand follows a Poisson distribution. Whenever the stock level reaches 5 dozens, a new order of 18 new dozens is placed for delivery at the beginning of the following week. Because of the nature of the item, all roses left at the end of the week are displaced of. Determine the following :

(i) The probability of placing an order in any one day of the week

(ii) The average number of dozen roses that will be discarded at the end of the week.

14. (a) Derive L_s, W_s, L_q and W_q for $(M/M/1)$: $(GD/\infty/\infty)$ queuing model.

Or

(b) Derive Pollaczek-Khintchine formula.

15. (a) Explain steepest ascent method.

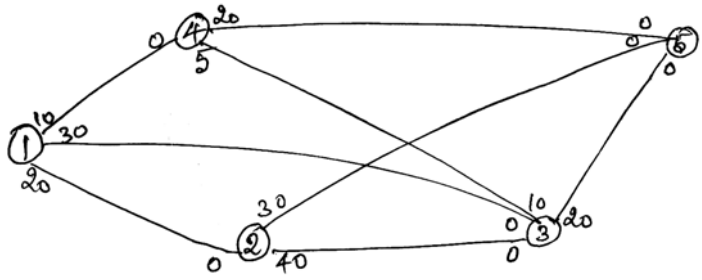
Or

(b) Enumerate the constrained algorithm.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Determine the maximum flow in the following network.



17. Neon lights on the U of A campus are replaced at the rate of 100 units per day. The physical plant orders the neon lights periodically. It costs \$100 to initiate a purchase order. A neon light kept in storage is estimated to cost about \$0.02 per day. The lead time between placing and receiving an order is 12 days. Determine the optimal inventory policy for ordering the neon lights.

18. Enumerate the pure birth and death models in queuing theory.

19. Visitors parking at Ozark College is limited to five spaces only. Cars making use of this space arrive according to a Poisson distribution at the rate of six cars per hour. Parking time is exponentially distributed with a mean of 30 minutes. Visitors who cannot find an empty space on arrival may temporarily wait inside the lot until a parked car leaves. That temporary space can hold only three cars. Other cars that cannot park or find a temporary waiting space must go elsewhere. Determine the following :

    (a) The probability, P_n of n cars in the system

    (b) The effective arrival rate for cars that actually use the lot

    (c) The average number of cars in the lot

    (d) The average time a car waits for a parking space inside the lot

    (e) The average number of occupied parking spaces.

20. Solve the following problem using Gradient method

    Maximize $f(x_1, x_2) = 4x_1 + 6x_2 - 2x_1^2 - 2x_1 x_2 - 2x_2^2$.

AFC–7742

Sub. Code
1MMA3E3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Mathematics

Elective — FUZZY MATHEMATICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Compute the scalar cardinality for the fuzzy set

   $c(x)=\dfrac{x}{x+1}$ for $x\in\{0,1,2.....10\}=x$.

2. Define support of fuzzy set. Give an example.

3. Define the intersection of two fuzzy sets. Give an example.

4. What is meant by sugeno class?

5. Define compatibility relation.

6. Using binary relations given by $M_1=\begin{pmatrix} .6 & .6 & 0 \\ 0 & .6 & .1 \\ 0 & .1 & 0 \end{pmatrix}$ and

   $M_2=\begin{pmatrix} 1 & 0 & .7 \\ 0 & 1 & 0 \\ .7 & 0 & 1 \end{pmatrix}$ find $M_1 * M_2$.

7. State the axioms of fuzzy measures.

8. What do you mean by plausibility measure?

9. What is a fuzzy constraint? Give an example.

10. Narrate the multi attributive decision making.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) For any $A \in \mathcal{F}(x)$, prove the following

(i) $^{\alpha}A = \bigcap_{\beta<\alpha} \beta_A = \bigcap_{\beta<\alpha} \beta^{+}A$.

(ii) $^{\alpha+}A = \bigcup_{\alpha<\beta} \beta_A = \bigcup_{\alpha<\beta} \beta^{+}A$.

Or

(b) State and prove third decomposition theorem.

12. (a) Show that every fuzzy complement has at most one equilibrium.

Or

(b) Show that $u(a,b) = \max(a,b)$ is the only continuous and idempotent fuzzy set union.

13. (a) Let $M_p = \begin{bmatrix} 0.3 & 0.4 & 0.5 \\ 0 & 1 & 0.1 \\ 0.2 & 0.6 & 0.7 \end{bmatrix}$ and

$$M_Q = \begin{bmatrix} 0.3 & 0.5 & 0.6 & 0.7 \\ 0 & 1 & 0.3 & 0.2 \\ 1 & 0.2 & 0.3 & 0.5 \end{bmatrix}$$

show that $(P\,o\,Q)^{-1} = Q^{-1}\,o\,P^{-1}$.

Or

(b) Explain the following types of fuzzy relations. Give an example for each :

(i) Reflexive.

(ii) Symmetric.

(iii) Transitive.

14. (a) Let a given finite body of evidence ($\mathcal{F}$, m) be nested. Prove that the associated belief and plausibility measures have the following properties for all $A, B \in P(x)$:

(i) $Bel\,(A \cap B) = \min[Bel(A), Bel(B)]$.

(ii) $Pl(A \cup B) = \max[Bel(A), Bel(B)]$.

Or

(b) Let $X = \{a, b, c, d\}$. Given belief measure $Bel(\{b\}) = 0.1$, $Bel(\{a, b\}) = 0.2$, $Bel(\{b, c\}) = 0.3$, $Bel(\{b, d\}) = 1$. Compute the corresponding basic assignment.

15. (a) Enumerate the vector maximum problem and its optimal compromise solution.

Or

(b) Narrate the fuzzy dynamic programming.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Consider the fuzzy sets A, B and C defined on the interval $X = [0,10]$ of real numbers by the $\mu_A(x) = \dfrac{x}{x+2}$, $\mu_B(x) = 2^{-x}$, $\mu_c(x) = \dfrac{1}{1 + 10(x-2)^2}$ and let $f(x) = x^2$ for all $x \in x$. Derive $f(A), f(B)$ and $f(C)$ using extension principle.

17. Show that fuzzy set operations of union, intersection, and continuous complement that satisfy the law of excluded middle and the law of contradiction are not idempotent or distributive.

18. (a) Narrate the procedure to solve fuzzy relation equations.

    (b) Solve the fuzzy relation equation.

$$P_0 \begin{bmatrix} 0.9 & 0.6 & 1 \\ 0.8 & 0.8 & 0.5 \\ 0.6 & 0.4 & 0.6 \end{bmatrix} = \begin{bmatrix} 0.6 & 0.6 & 0.5 \end{bmatrix}.$$

19. Let $x = \{a, b, c, d, e, f, g\}$ and $y = N_7$. using joint probability distributions on Xxy, given in terms of the matrix.

	1	2	3	4	5	6	7
a	0.08	0	0.02	0	0	0.01	0
b	0	0.05	0	0	0.05	0	0
c	0	0	0	0	0	0	0.03
d	0.03	0	0	0.3	0	0	0
e	0	0	0.01	0.01	0.2	0.03	0
f	0	0.05	0	0	0	0.1	0
g	0	0	0	0.02	0	0.01	0

Compute :

(a) Marginal probabilities.

(b) Both conditional probabilities.

(c) Hypothetical joint probability distribution based on the assumption of non-interaction.

20. Consider the following problem :

Maximize $z = 2x_1 + x_2$

Subject to $x_1 \le 3$

$x_1 + x_2 \le 4$

$5x_1 + x_2 \le 3$

$x_1, x_2 \ge 0$

Determine the optimal solution.

AFC–7743

Sub. Code
1MMA4C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fourth Semester

Mathematics

FUNCTIONAL ANALYSIS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. State Jensen's inequality.

2. Show that $\mathbf{R}$ is a finite dimensional normed linear space.

3. Let X be a normed space over K, $f \in X^1$ and $f \neq 0$. Let $a \in X$ with $f(a) = 1$ and $r > 0$. Prove that $U(a,r) \cap Z(f) = \phi$ if and only if, $\|f\| \leq \frac{1}{r}$.

4. Write a note on Banach limit.

5. State uniform boundedness principle.

6. Define dual of a normed space. Give an example.

7. Define the spectrum of a Bounded operator.

8. Does the Riesz representation theorem hold for incomplete inner product space? Justify.

9. Define Weak* Convergence of a sequence.

10. Define orthogonal projection.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Show that a compact subset of a normed space is closed and bounded.

Or

(b) Prove that the linear functional f and C defined by $f(x) = \lim_{\bar{j} \to \infty} x(\bar{j}), x \in C$, is continuous and $\|f\| = 1$.

12. (a) Prove that the dual X' of every normed space X is a Banach space.

Or

(b) For a normed space Y, prove that $BL(X,Y) = \{0\}$ if and only if $Y = \{0\}$.

13. (a) State and prove Resonance theorem.

Or

(b) Let X be an infinite dimensional normed space. If Y is a non-zero normed space, then prove that there is some $F : X \to Y$ which is linear but non-closed.

14. (a) Let X be a normed space. Prove that if X' is separable, then so is X.

Or

(b) Let X and Y be Banach spaces and $F \in BL(X,Y)$ prove that $R(F) = Y$ if, and only if F' is bounded below.

15. (a) State and prove parallelogram law.

Or

(b) State and prove projection theorem.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Let X denote a subspace of $B(T)$ with the sup norm, $1 \in X$ and f be a linear functional on X. Prove that if f is continuous and $\|f\| = f(1)$, then f is positive. Conversely, if $\operatorname{Re} x \in X$ whenever $x \in X$ and if f is positive, then f is continuous and $\|f\| = f(1)$.

17. Prove that a normed space X is a Banach space if and only if every absolute summable series of elements in X is summable in X.

18. State and prove closed graph theorem.

19. State and prove Riesz representation theorem for $C[a,b]$.

20. (a) State and prove Bessel's inequality.

(b) State and prove Schwarz inequality.

AFC–7744

Sub. Code
1MMA4C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fourth Semester

Mathematics

NUMBER THEORY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Show that these are infinitely many primes.

2. If a prime number p does not divide 'a' , then show that $(p,a)=1$.

3. Define the Euler totient function.

4. Define Riemann zeta function.

5. State the average orders of $\mu(n)$ and $\lambda(n)$.

6. If 'n' is a positive integer and x is any real numbers, then prove that $\left[\frac{x}{n}\right]=\left[\frac{[x]}{n}\right]$.

7. State Little fermat theorem.

8. What are the solutions of the congruences $x^2 \equiv 1 \pmod 8$?

9. What are the quadratic residues and non-residuces mod 13?

10. Write down the values of $\left(\frac{2}{17}\right)$ and $\left(\frac{2}{19}\right)$.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) on (b).

11. (a) State and prove fundamental theorem of arithmetic.

Or

(b) State and prove the division algorithm.

12. (a) For $n \geq 1$, prove that $\phi(n) = n\prod_{p/n}\left(1-\frac{1}{\beta}\right)$.

Or

(b) For $n \geq 1$, prove that $\phi(n) = \sum_{d/n}\mu(d)\frac{n}{d}$.

13. (a) Prove the Legendre's identity.

Or

(b) For $x > 1$, prove that $\sum_{n\leq x}\phi(n) = \frac{3x^2}{\pi^2} + 0(x\log x)$.

14. (a) Prove that $5n^3 + 7n^5 \equiv 0 \pmod{12}$ for all $n \in z$.

Or

(b) State and prove Euler-Fermat theorem.

15. (a) Prove that the Legendre's symbol (n/p) is completely multiplicative function of n.

Or

(b) If p is an odd positive integer, then prove that $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. State and prove Euclidean algorithm.

17. If both g and $f * g$ are multiplicative, then prove that f is also multiplicative.

18. State and prove Euler's summation formula.

19. State and prove Wolstenholme's theorem.

20. State and prove Gauss Lemma.

AFC–7745

Sub. Code
1MMA4C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fourth Semester

Mathematics

TOPOLOGY — II

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Is the real line **R** is locally compact? Justify your answer.

2. State the countable intersection property.

3. When do we say that the given two compactification are equivalent?

4. Define the stone–cech compactification.

5. What is meant by closed refinement?

6. Define G_S – set. Give an example.

7. Is the space $(-1,1)$ in **R** complete? Justify.

8. Define the point open topology.

9. What do you mean by evaluation map?

10. State Ascoli's theorem.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Show that $[0,1]^w$ is not locally compact in the uniform topology.

Or

(b) Let X be locally compact Hausdorff and A be a subspace of X. If A is closed in X or open in X, then prove that A is locally compact.

12. (a) Show that a product of completely regular spaces is completely regular.

Or

(b) If X is completely regular, then prove that X can be imbedded in $[0,1]^J$ for some J.

13. (a) Show that if X has a countable basis, a collection A of subsets of X is countably locally finite if and only if it is countable.

Or

(b) Find a nondiscrete space that has a countably locally finite basis but does not have a countable basis.

14. (a) Prove that there is a metric for the product space R^w relative to which R^w is complete.

Or

(b) Let *(X,d)* be a metric space. Prove that there is an isometric imbedding of X into a complete metric space.

15. (a) Define the following terms:

(i) Compact open topology.

(ii) σ–compact

(iii) Baire space

Or

(b) Show that every locally compact Hausdorff space is a Baire space.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. State and prove the Tychonoff theorem.

17. Let X be a completely regular space. Prove that there exists a compactification Y of X having the property that every bounded continuous map $f : X \to R$ extends uniquely to a continuous map of Y into R.

18. State and prove the Nagata–Smirnov metrization theorem.

19. Let $I = [0,1]$. Prove that there exists a continuous map $f : I \to I^2$ whose image fills up the entire square I^2.

20. Show that a metric space (X, d) is compact if and only if it is complete and totally bounded.

AFC–7746

Sub. Code
1MMA4C4

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fourth Semester

Mathematics

DISCRETE MATHEMATICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Let $S = \{1, -1, 0\}$. Show that the addition '+' is not binary on S but the multiplication '×' is a binary operation on S.

2. Define a monoid with an example.

3. Write down the negation of the statement "Our school produced 100% result in the last year's final".

4. Show that $Q \Rightarrow P \rightarrow Q$.

5. Define an open statement.

6. Write the statement "No men are giants" in symbolic form.

7. Define a least upper bound (l.u.b) with an example.

8. Show that every distributive lattice is modular.

9. Define a Boolean homomorphism.

10. Expand the following function into the canonical sum-of-products forms: $f(x, y, z) = xy + y\bar{z}$.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Complete the following table, so that the binary operation $*$ is associative.

*	a	b	c	d
a	a	b	c	d
b	b	a	c	d
c				
d	d	c	c	d

Or

(b) Show that the property of idempotency is preserved under a semi group homomorphism.

12. (a) Construct the truth table of the formula $(\neg P \vee Q) \wedge (\neg Q \vee P)$.

Or

(b) Find a disjunctive normal form of $(q \vee (p \wedge r)) \wedge \neg((p \vee r) \wedge q)$.

13. (a) Show that $R \to S$ can be derived from the premises $P \to (Q \to S)$, $\neg R \vee P$, and Q.

Or

(b) Verify the validity of the following argument:

All men are mortal

Socrates is a man

Therefore Socrates is a mortal.

14. (a) Show that, in any lattice $(L, \leq)$, the operations $\vee$ and $\wedge$ are isotone.

Or

(b) If G is a group, then prove that the set of all normal subgroups of G forms a modular lattice.

15. (a) Let B be the Boolean algebra $\{0, 1\}$ with the usual $\wedge, \vee$, operations and A be any given finite Boolean algebra. Let $p_1 q \in p_n$. If $p_B = q_B$ then prove that $p_A = q_A$.

Or

(b) Write down the minimization algorithm for Boolean polynomials.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Let T be the set of all even integers, show that the semigroups $(Z, +)$ and $(T, +)$ are isomorphic.

17. (a) Obtain the principal disjunctive normal form of $P \to ((P \to Q) \wedge \neg(\neg Q \vee \neg P))$.

(b) Obtain a disjunctive normal form of $\neg(P \vee Q) \leftrightarrow (P \wedge Q)$.

18. (a) Using indirect method of proof, derive $P \to \neg S$ from $P \to Q \vee R, Q \to \neg P, S \to \neg R, P$.

(b) Show that $(\forall x)(P(x) \wedge Q(x)) \leftrightarrow ((\forall x)(P(x) \wedge (\forall x)(Q(x))))$ is a logically valid statement.

19. In a distributive lattice, prove that the following are equivalent

(a) $a \wedge b \leq x \leq a \vee b$

(b) $x = (a \wedge x) \vee (b \wedge x) \vee (a \wedge b)$.

20. Simplify

$f(a, b, c, d, e) = \sum (0,1,3,8,9,13,14,15,16,17,19,24,25,27,31)$.

AFC–7727

Sub. Code
1MMA1C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Mathematics

ALGEBRA – I

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. State Sylow's theorem for Abelian groups.

2. Show that a group of order 36 is not simple.

3. Define internal direct product of groups.

4. Prove that $N(a)$ is a subgroup of G.

5. Define a ring homomorphism.

6. If $\phi : R \to R'$ is a ring homomorphism with Kernal $I(\phi)$, then prove that $I(\phi)$ is a subgroup of R under addition.

7. If U and V are ideals of R, then prove that $U + V$ is also an ideal of R.

8. Define over-ring.

9. Define content of the polynomial and give an example.

10. Let R be an Euclidean ring. Suppose that for $a,b,c \in R, a|bc$ but $(a,b)=1$ then prove that $a|c$.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Prove that if H and K are finite subgroups of G of order $O(H)$ and $O(K)$ respectively, then $O(HK)=\dfrac{O(H)\cdot O(K)}{O(H\cap K)}$.

Or

(b) Prove that every permutation is the product of its cycles.

12. (a) Let G be a group and suppose that G is the internal direct product of $N_1, N_2, ...N_n$. Let $T = N_1 \times N_2 \times ... \times N_n$ then prove that G and T are isomorphic.

Or

(b) Prove that $n(k)=1+p+p^2+...+p^{k-1}$.

13. (a) If $\phi : R \to R'$ is a ring homomorphism with Kernal $I(\phi)$, then prove that if $a \in I(\phi)$ and $r \in R$ then both ar and ra are in $I(\phi)$.

Or

(b) If $a,b \in R$, where R is a ring then prove that $(a+b)^2 = a^2 + ab + ba + b^2$ where $x^2 = x \cdot x$.

14. (a) Let R be the ring of all real valued continuous functions on the closed unit interval and let $M = \{f(x) \in R : f(\frac{1}{2}) = 0\}$ then prove that M is a maximal ideal of R.

Or

(b) If R is a commutative ring with unit element and M is an ideal of R, then prove that M is a maximal ideal of R if and only if R/M is a field.

15. (a) If p is a prime number of the form $4n+1$, then solve the congruences $x^2 \equiv (-1)(\text{mod } p)$.

Or

(b) Prove that the cyclotonic polynomial $1 + x + x^2 + \ldots + x^{p-1}$ where p is a prime number is irreducible over the field of rational numbers.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. State and prove Cayley's theorem.

17. State and prove third part of Sylow's theorem.

18. Prove that the homomorphism $\phi : R \to R'$ is an isomorphism if and only if $I(\phi) = \{0\}$.

19. If U is an ideal of a ring R then prove that R/U is a ring and is a homomorphic image of R.

20. Prove that $J[i]$ is an Euclidean ring.